

BCS508 – ENVIRONMENTAL STUDIES & E-WASTE MANAGEMENT

MODULE-01 | 100 MCQ

1. The term Environment has been derived from the French word which means:
 - A. Environ
 - B. Oikos
 - C. Geo
 - D. Aqua
2. The term environment was introduced by:
 - A. Jacob Van Verkul
 - B. Tyler Miller
 - C. Gilbertson
 - D. Eugene P Odum
3. The word Environment is derived from:
 - A. Greek
 - B. French
 - C. Spanish
 - D. English
4. Environment is the life support system that includes:
 - A. Air
 - B. Water
 - C. Land
 - D. All of the above
5. The objective of environmental education is:
 - A. Raise consciousness about environmental conditions
 - B. To teach environmentally appropriate behavior
 - C. Create an environmental ethic
 - D. All of the above
6. Which conceptual sphere of the environment has the least storage capacity for matter?
 - A. Atmosphere
 - B. Lithosphere
 - C. Hydrosphere
 - D. Biosphere
7. Which components of the environment are effective transporters of matter?
 - A. Atmosphere and Hydrosphere
 - B. Atmosphere and Lithosphere
 - C. Hydrosphere and Lithosphere
 - D. Biosphere and Lithosphere
8. Amount of CO₂ content in atmospheric air is normally:
 - A. 0.21%
 - B. 0.416%
 - C. 0.318%
 - D. 0.428%
9. Among fresh water availability on earth, the percentage of groundwater is about:
 - A. 0.2%
 - B. 0.5%

C. 0.8%

D. 1.0%

10. Percentage of fresh water available on earth is:

A. 2.8%

B. 2.2%

C. 0.6%

D. 2.15%

11. About what percentage of the earth is covered by water?

A. 53%

B. 19%

C. 71%

D. 33%

12. Among fresh water available on earth, the percentage of surface water is about:

A. 50%

B. 10%

C. 5%

D. < 1%

13. The rain forms represented by snow, dew, fog and mist are known as:

A. Dry deposition

B. Wet deposition

C. Acid rain

D. None

14. The temperature in the stratosphere ranges between:

A. 25°C to 55°C

B. -56°C to -2°C

C. 10°C to -5°C

D. None

15. Troposphere mainly consists of:

A. CO₂

B. H₂

C. O₃

D. O₂ and N₂

16. The volume of nitrogen present in the troposphere is:

A. 85%

B. 78%

C. 21%

D. 5%

17. Inverse condition is the:

A. Increase in temperature with altitude

B. Decrease in temperature with altitude

C. Increase in temperature with latitude

D. Decrease in temperature with latitude

18. Carbon content is higher in:

A. Soil

B. Atmosphere

C. Water

D. Living matter

19. Atmosphere consists of 79 per cent Nitrogen and 21 per cent Oxygen by:

- A. Volume**
- B. Weight
- C. Density
- D. All the three

20. The major atmospheric gas layer in stratosphere is:

- A. Hydrogen
- B. Carbon dioxide
- C. Ozone**
- D. Helium

21. UV-C radiation lies in the region:

- A. 315 to 400 nm
- B. < 280 nm**
- C. > 280 nm
- D. 280 to 315 nm

22. Which atmospheric sphere is closest to the earth surface?

- A. Troposphere**
- B. Stratosphere
- C. Mesosphere
- D. Exosphere

23. The largest reservoir of nitrogen in our planet is:

- A. Oceans
- B. Atmosphere**
- C. Biosphere
- D. Fossil fuels

24. Which among the components is termed as a life zone on the earth?

- A. Atmosphere
- B. Hydrosphere
- C. Biosphere**
- D. Stratosphere

25. Biosphere is:

- A. The solid shell of inorganic materials or the surface of the earth
- B. The thin shell of organic matter on the surface of earth comprising of all the living things**
- C. The sphere which occupies the maximum volume of all the spheres
- D. All of the above

26. Loss of water content through plants into atmosphere is called:

- A. Evaporation
- B. Transpiration**
- C. Vaporization
- D. Hydraulic cycle

27. Which of the following is absorbed by green plants from the atmosphere?

- A. Carbon dioxide**
- B. Water
- C. Nutrients
- D. All of above

28. The word Ecology is proposed by:

- A. Ernst Haeckel**
- B. Helena Curtis
- C. Charles Southwick
- D. Charles Alton

29. The science that deals with the relationship of various organisms with their environment is known as:

- A. Economics
- B. Geology
- C. Ecology**
- D. Anthropology

30. Habitat refers to:

- A. Physical conditions of the place where organisms live
- B. Chemical conditions of the place where organisms live
- C. Both a & b**
- D. None of a or b

31. Which of the following is a biotic component of an ecosystem?

- A. Fungi**
- B. Solar light
- C. Temperature
- D. Humidity

32. In an ecosystem, the flow of energy is:

- A. Bi-directional
- B. Cyclic
- C. Unidirectional**
- D. Multidirectional

33. Which pyramid is always upright?

- A. Energy**
- B. Biomass
- C. Producers
- D. Food chain

34. In complex ecosystem existence of species diversity is:

- A. Poor
- B. High**
- C. Medium
- D. None

35. The organisms who directly feed on producers are called:

- A. Herbivores**
- B. Carnivores
- C. Decomposers
- D. Saprophytes

36. The sequence of eating and being eaten in an ecosystem is called:

- A. Food Chain**
- B. Carbon cycle
- C. Hydrological cycle
- D. Anthropogenic system

37. Which of the below is a producer in an ecosystem?

- A. Plants and some bacteria capable of producing their own food**
- B. Animals
- C. Human beings
- D. Fish

38. Which of the following statements is false?

- A. Inorganic nutrients are recycled in an ecosystem
- B. Energy flows through the ecosystem in the form of carbon bonds
- C. Energy is recycled in an ecosystem**
- D. Respiration process releases energy

39. In aquatic ecosystem phytoplankton can be considered as a:

- A. Consumer
- B. Producer**
- C. Saprophytic organisms
- D. Macro consumer

40. The basic requirements of human beings are provided by:

- A. Industrialization
- B. Agriculture
- C. Nature**
- D. Urbanization

41. In an ecosystem biological cycling of materials is maintained by:

- A. Producer
- B. Consumer
- C. Decomposer
- D. All of the above**

42. Organisms which feed directly or indirectly on producers are called:

- A. Prey
- B. Consumers**
- C. Decomposers
- D. Detritus

43. The primary producers in a forest ecosystem are:

- A. Chlorophyll containing trees and plants**
- B. Herbivores
- C. Carnivores
- D. Bacteria and other micro-organisms

44. Abiotic components include:

- A. Soil
- B. Temperatures
- C. Water
- D. All the above**

45. Which of the following is true?

- A. Green plants are self nourishing**
- B. Producers depends on consumers
- C. Biotic components includes all nonliving components
- D. Herbivores depend on Carnivores

46. Primary consumer is:

- A. Herbivores**
- B. Carnivores
- C. Macro consumers
- D. Omnivores

47. A predator is:

- A. An animal that is fed upon
- B. An animal that feeds upon another animal**
- C. Animal that feeds upon both plants and animals
- D. A primary consumer

48. Access to food is mainly determined by:

- A. Household income
- B. Food assistance programs
- C. Human resources**
- D. Society/Community

49. Which among the following is a climatic factor?

- A. Pressure
- B. Humidity
- C. Temperature
- D. All of the above**

50. Which of the following is the terrestrial ecosystem?

- A. Forest
- B. Grass land
- C. Desert
- D. All of the above**

51. Ecological pyramids are studies of:

- A. Pyramid of numbers
- B. Pyramid of biomass
- C. Pyramid of energy
- D. All of the above**

52. A trophic level of organism represents:

- A. An organism's position in a biomass
- B. An organism's position in food chain**
- C. Group of organisms in the food chain
- D. An organism's position in an ecosystem

53. Factors responsible for balanced ecosystem are:

- A. Balance between predator and prey
- B. Balance between vegetation, herbivores and carnivorous
- C. Balance between competing species and biotic factors
- D. All of the above**

54. A food web consists of:

- A. A portion of a food chain
- B. An organism's position in a food chain
- C. Interlocking food chains**
- D. A set of similar consumers

55. Which of the following statements are true?

- A. Man is not dependent on nature
- B. Resources are unlimited, so one can use them as per ones wish
- C. Energy can be converted from one form to another, but some percentage is lost into the environment**
- D. Matter can be generated afresh; it need not be recycled or reused

56. Which of the following is not a desirable chemical in a good habitat?

- A. Oxygen
- B. Carbon dioxide
- C. Sulphur dioxide**
- D. Nutrients

57. An ecosystem is a region in which:

- A. Dead organism interact with their environment
- B. Living organism do not interact with their environment

C. Living organism interact with their environment

D. All these

58. The perfect equilibrium existing in the biosphere between the various organisms is known as:

A. Environmental balance

B. Ecological Balance

C. Ecological cycle

D. None of these

59. The term ecosystem was first proposed by:

A. Jacob Van Verkul

B. A.G. Tansley

C. Costana

D. Marie Gibbs

60. The two major components of ecosystem are:

A. Adiabatic & isotropic

B. Ecological & climatological

C. Cyclic & biologic

D. Abiotic & biotic

61. Biotic components include:

A. All living organisms

B. Water, minerals and gases

C. Self-nourishing green plants

D. Light, temperature, etc

62. Food chain is divided into _____basic categories:

A. Four

B. Three

C. Five

D. Seven

63. The transfer of energy and nutrients from one feeding group of organisms to another in a series is called:

A. Energy chain

B. Food chain

C. Balanced ecosystem

D. Food web

64. A simple food chain starts with:

A. Green plants

B. Waste of organisms and dead organisms

C. Both of these

D. None of these

65. In a food chain humans are:

A. Primary consumers

B. Secondary consumers

C. Primary and secondary consumers

D. Producers

66. Tendency of pollutants to become concentrated in successive trophic levels is known as:

A. Bioremediation

B. Biomagnifications

C. Biopiracy

D. Biorhythm

67. Biodiversity is a measure of variation at the ____level.

- A. Genetic
- B. Species
- C. Ecosystem
- D. All of the above**

68. Terrestrial biodiversity is usually greater near the:

- A. North pole
- B. South pole
- C. Equator**
- D. Tropic of cancer

69. The age of earth is about ____billion years.

- A. 3.04
- B. 3.54
- C. 4.05
- D. 4.54**

70. Following is usually measured at the species diversity level:

- A. Ecological diversity
- B. Taxonomic diversity**
- C. Morphological diversity
- D. Functional diversity

71. Diversity of all living things depends on:

- A. Geography
- B. Presence of other species
- C. Soils
- D. All of the above**

72. A biodiversity hotspot:

- A. Is a region with a high level of endemic species
- B. Has experienced great habitat loss
- C. Both (A) and (B)**
- D. None of the above

73. Biodiversity is directly involved in:

- A. Water purification
- B. Recycling nutrients
- C. Providing fertile soil
- D. All of the above**

74. The most studied group is:

- A. Birds**
- B. Mammals
- C. Both (A) and (B)
- D. Fishes

75. Following step(s) can conserve the forest cover:

- A. Prevent forest fire
- B. Prevention of overgrazing by cattle
- C. Hunting and poaching should be banned
- D. All of the above**

76. In which of the following place can we find the cold deserts?

- A. Bangalore
- B. Chennai

C. Himalaya

D. Rajasthan

77. How can desert ecosystems be conserved?

A. By minimizing the human activity

B. By pouring water to desert area

C. By deforestation

D. By killing organisms

78. Which kind of soil can be found on the surface of Thar desert?

A. Rocky

B. Moist

C. Fertile

D. Aeolian

79. Which sphere contains most of the earth's water?

A. Atmosphere

B. Hydrosphere

C. Lithosphere

D. Biosphere

80. Which sphere represents the solid part of the earth?

A. Hydrosphere

B. Atmosphere

C. Lithosphere

D. Biosphere

81. Which atmospheric layer is above the troposphere?

A. Stratosphere

B. Exosphere

C. Hydrosphere

D. Lithosphere

82. The ozone layer is mainly associated with the:

A. Troposphere

B. Stratosphere

C. Mesosphere

D. Exosphere

83. Which gas is present at approximately 21% of the atmosphere by volume?

A. Nitrogen

B. Oxygen

C. Carbon dioxide

D. Ozone

84. Which gas is present at approximately 78% of the atmosphere by volume?

A. Oxygen

B. Nitrogen

C. Carbon dioxide

D. Hydrogen

85. Which of the following is a living component of an ecosystem?

A. Temperature

B. Water

C. Fungi

D. Solar light

86. Which of the following is a non-living component of an ecosystem?

- A. Fungi
- B. Plants
- C. Temperature**
- D. Animals

87. Which trophic level is occupied by producers?

- A. First**
- B. Second
- C. Third
- D. Fourth

88. Herbivores generally occupy the:

- A. Producer level
- B. Primary consumer level**
- C. Secondary consumer level
- D. Decomposer level

89. Carnivores that feed on herbivores are generally:

- A. Primary consumers
- B. Secondary consumers**
- C. Producers
- D. Decomposers

90. Which process returns nutrients to the ecosystem from dead organic matter?

- A. Decomposition**
- B. Photosynthesis
- C. Transpiration
- D. Evaporation

91. Which of the following best represents a food web?

- A. A single feeding relationship
- B. Interlocking food chains**
- C. Only producers
- D. Only decomposers

92. Which ecological pyramid represents energy available at successive trophic levels?

- A. Pyramid of numbers
- B. Pyramid of biomass
- C. Pyramid of energy**
- D. Pyramid of species

93. What happens to energy as it passes from one trophic level to the next?

- A. It is completely recycled
- B. Some energy is lost to the environment**
- C. It increases at every level
- D. It becomes matter

94. A habitat provides organisms with:

- A. Only food
- B. Only water
- C. Physical and chemical conditions for living**
- D. Only sunlight

95. Which practice helps conserve biodiversity?

- A. Habitat destruction
- B. Poaching
- C. Protection of habitats**

D. Overgrazing

96. Which action helps conserve forest cover?

A. Uncontrolled logging

B. Prevention of forest fires

C. Increasing overgrazing

D. Encouraging poaching

97. Which region is generally associated with high terrestrial biodiversity?

A. Polar regions

B. Equatorial regions

C. Permanent ice sheets

D. High-altitude glaciers only

98. Which of the following is an example of a terrestrial ecosystem?

A. Pond

B. Lake

C. Forest

D. Ocean

99. Which of the following is an example of an aquatic ecosystem?

A. Desert

B. Grassland

C. Forest

D. Pond

100. Which term describes the variety of life at genetic, species and ecosystem levels?

A. Biodiversity

B. Biomagnification

C. Transpiration

D. Deposition